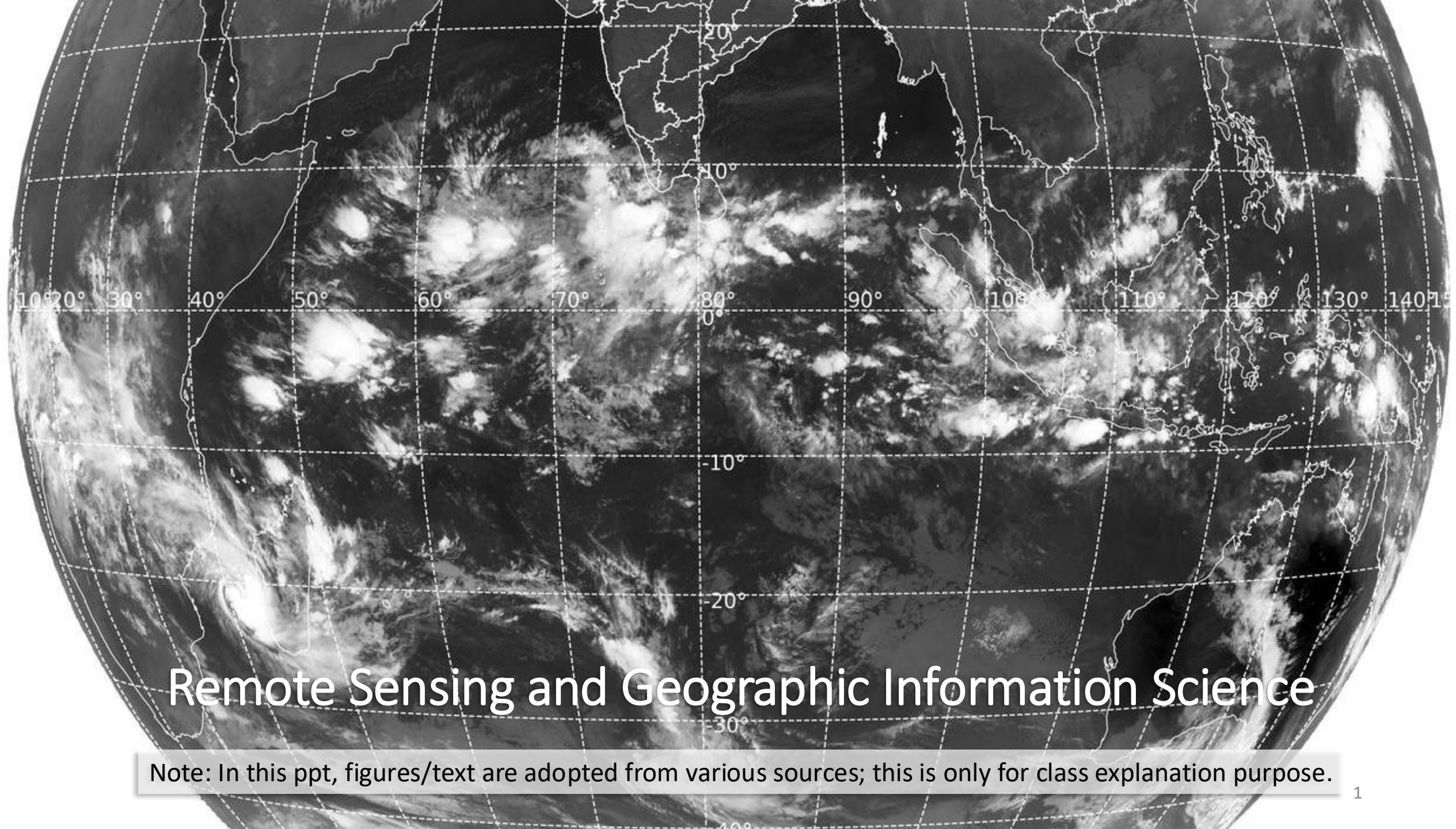




Remote Sensing and Geographic Information Science

Note: In this ppt, figures/text are adopted from various sources; this is only for class explanation purpose.

IMAGE PRE PROCESSING

- Pre processing refers to those operations which are preliminary (basis) for the principal analysis
- Pre-processing mainly deals with
 - Image restoration – Obtaining a clean/correct image from a noisy/corrupt image
 - Image rectification -- transformation process to project two-or-more images onto a common image plane
- Pre processing Operations:
 - 1) Radiometric Pre processing
 - To adjust digital values (DNs) for effects of hazy atmosphere
 - 2) Geometric Pre processing
 - To bring an image into registration with a map or another image
- Once Pre processing (corrections) is done, the data (image) can be subjected to primary analysis including:
 - 1) Image Classification ; 2) Change Detection ; 3) Land Use Land Cover

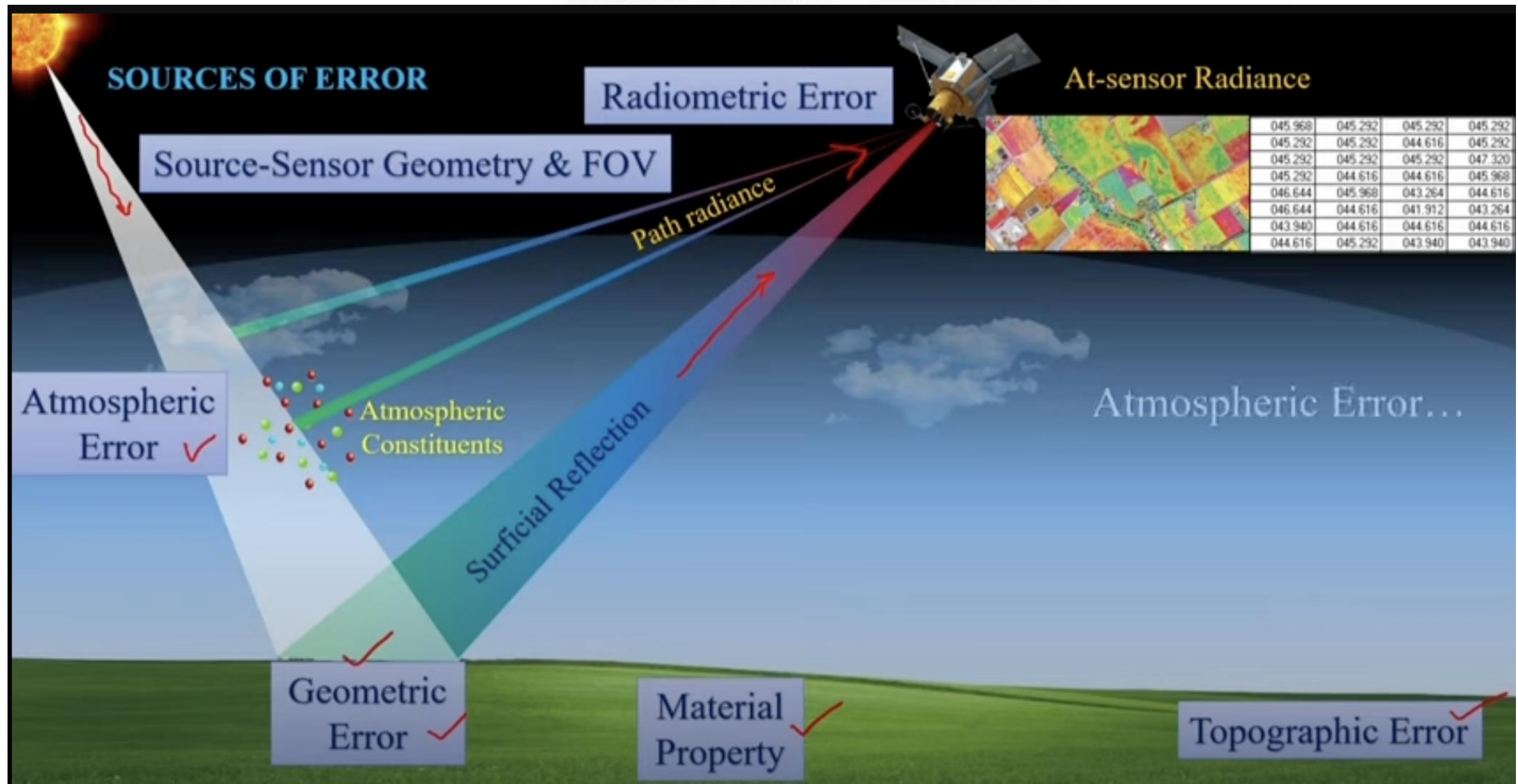


IMAGE PRE PROCESSING

- Preprocessing of satellite images prior to image classification and change detection is essential.
- Preprocessing commonly comprises a series of sequential operations, including atmospheric correction or normalization, image registration, geometric correction, and masking (e.g., for clouds, water, irrelevant features)
- Preprocessing is intended to correct for sensor- and platform-specific radiometric and geometric distortions of data
- Although the effects of the atmosphere upon remotely sensed data are not considered errors, since they are part of the signal received by the sensing device, consideration of these effects is important
- Geometric rectification of the imagery resamples or changes the pixel grid to fit that of a map projection or another reference image
- This becomes especially important when scene to scene comparisons of individual pixels in applications such as change detection are being sought

RADIOMETRIC PRE PROCESSING

- Radiometric correction is done to reduce or correct errors in the digital numbers of images
- This process improves the interpretability and quality of remote sensed data
- Radiometric calibration and correction are particularly important when comparing data sets over a multiple time periods
- Error in energy sensed by the sensor onboard can be due to: (Already discussed)
 - Sun's azimuth and elevation
 - Atmospheric conditions

SOME COMMON DEFINITIONS

Image Registration

The process of transforming different sets of data (multiple photographs, data from different sensors, times, depths, or viewpoints) into one coordinate system.

Image Restoration

The operation of taking a corrupt/noisy image (motion blur, noise) and estimating the clean, original image

Image Rectification

Transformation process used to project images onto a common image plane

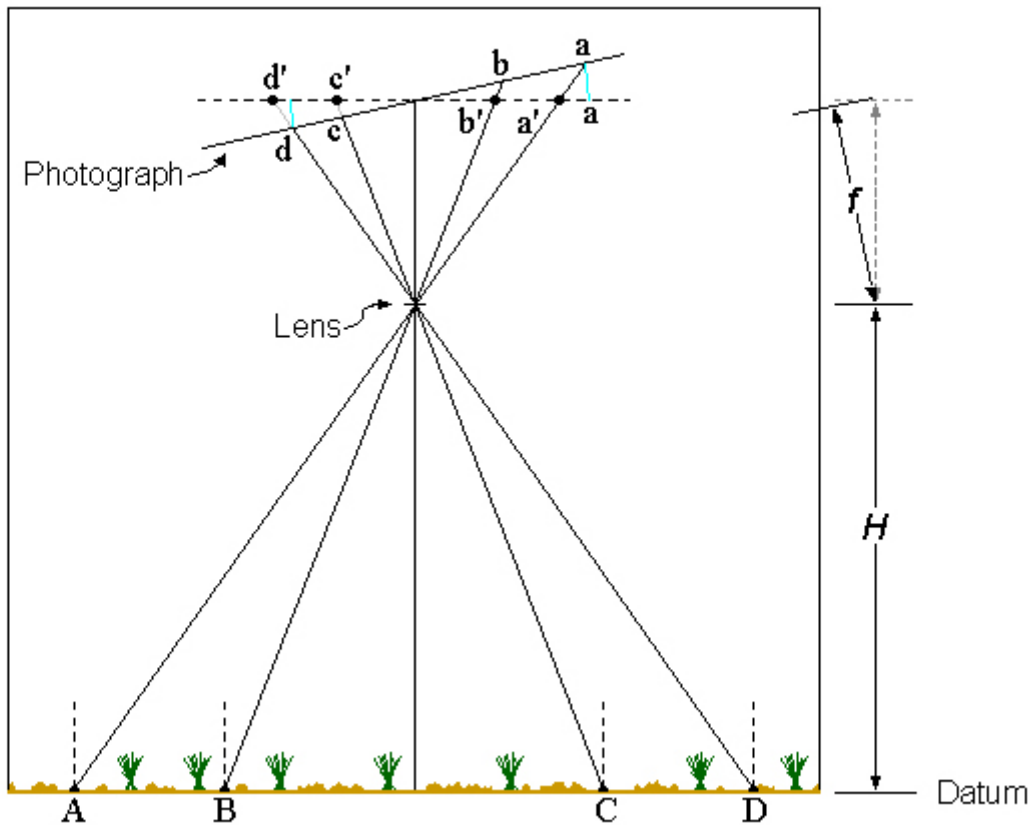
Orthorectification

process of removing the effects of image perspective (tilt) and relief (terrain) effects for the purpose of creating a planimetrically correct image

Image Mosaicing

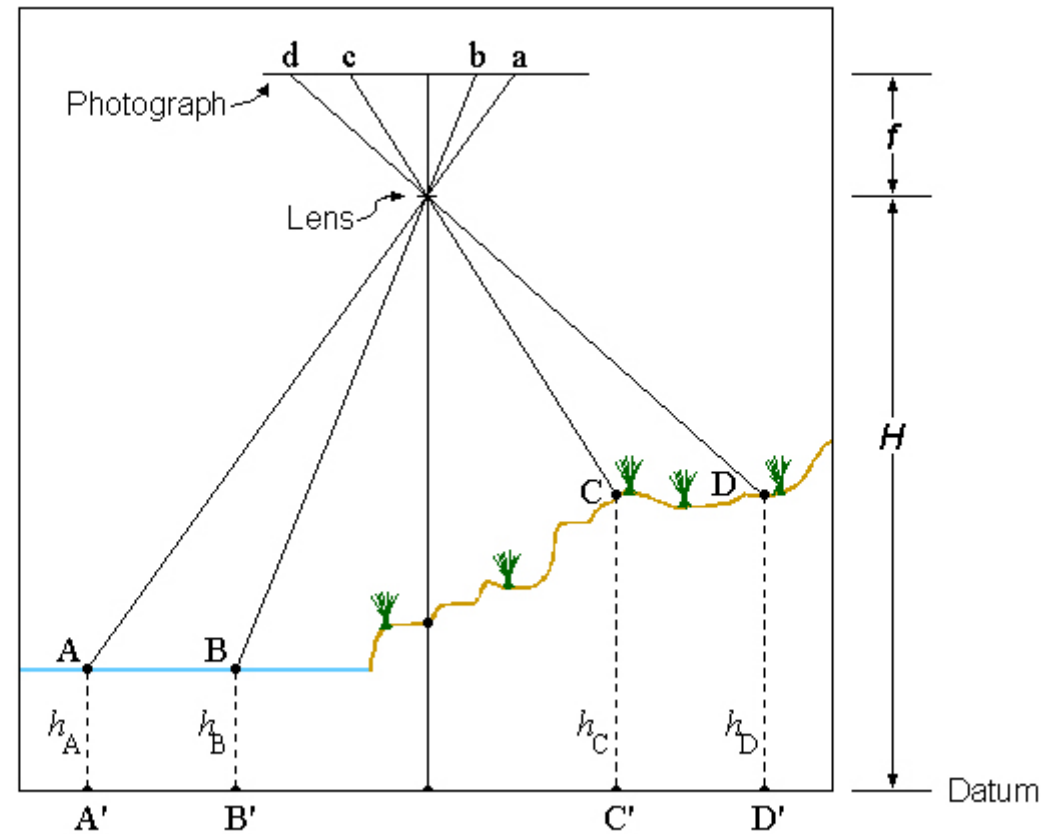
The process of assembling a series of images and joining them together to form a continuous seamless photographic representation

TILT DISPLACEMENT



Tilt displacement is the shift in an object's image position on a tilted photo from its theoretical position on a truly vertical photo

RELIEF DISPLACEMENT



Relief displacement is the shift in an object's image position caused by its elevation above a particular datum

DATA PRODUCTS – BASED ON PROCESSING

- 1) **Level 0** imagery is raw instrument data, just as they were collected at the sensor. Most agencies will not distribute Level 0 imagery
- 2) **Level 1A** data have been corrected for detector variations within the sensor.
Number of detectors depends on the instrument design (Landsat TM: 16). Satellite operators try to equalize the detector responses through a series of data calibrations involving pre-launch values and in-flight measurements. Level 1A processing involves applying inter-detector equalizations (*radiometric correction*)

DATA PRODUCTS – BASED ON PROCESSING

3) Level 1B involves the geometry of the image product.

Used to rectify distortions in the geometry of the imagery such as misaligned scan lines and non-uniform pixel sizes.

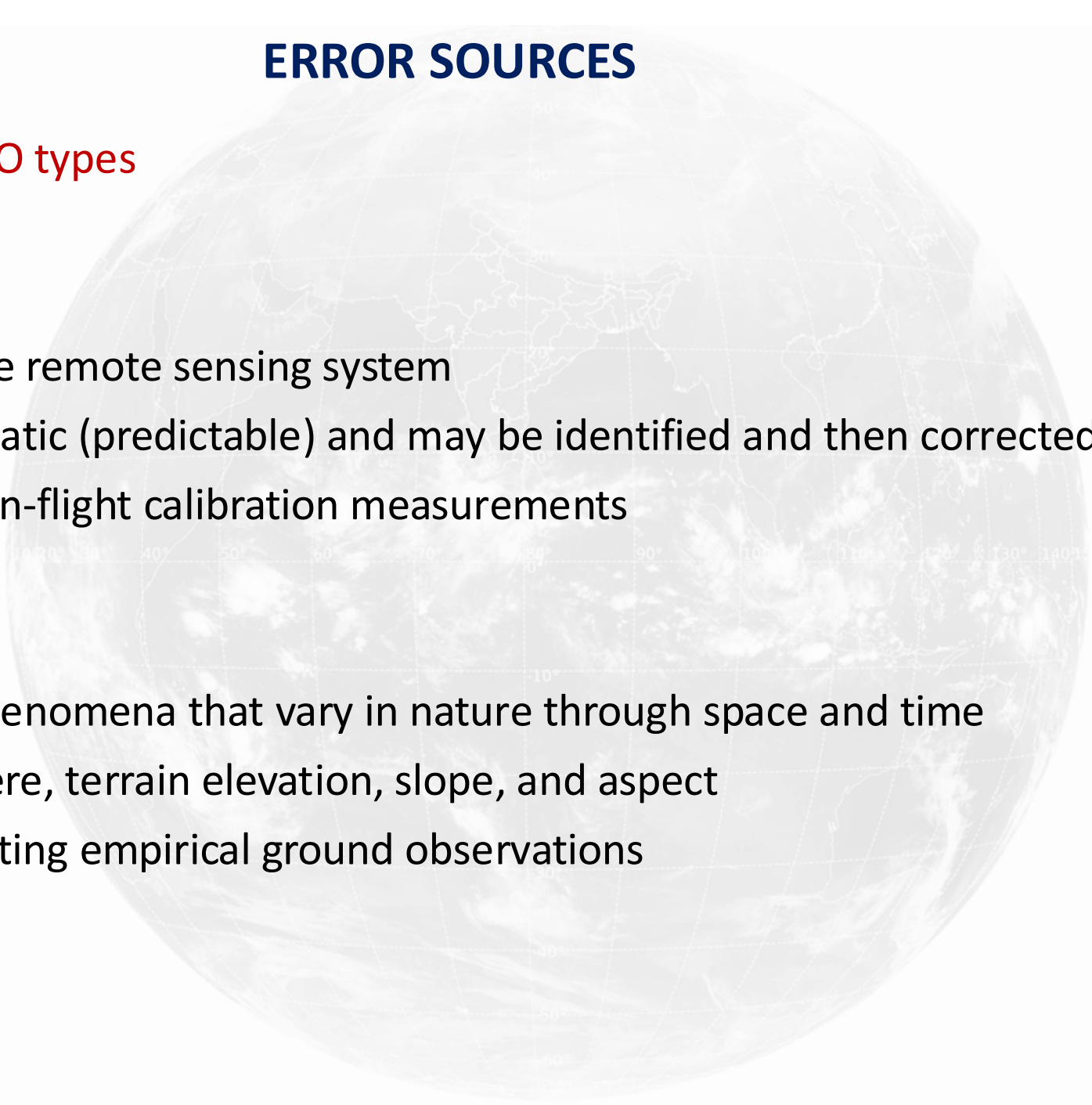
All of these distortions are predictable and measurable systematic corrections can be applied to the imagery to improve its geometric qualities

4) Level 2A images have been systematically mapped into a standard cartographic map projection based on a prediction of where the satellite was when the image was acquired. However, these images still have expected location errors of more than 100 m (can be used with GIS)

DATA PRODUCTS – BASED ON PROCESSING

- 5) **Level 2B** In this, image analyst "registers" the image to an existing base map by selecting pairs of well-defined points (known as *ground control points*) from both the image and the map. The position accuracy of Level 2B images generally matches the spatial resolution of the original data
- 6) **Level 3A** In addition to manually locating ground control points (as in Level 2B), a digital elevation model (DEM) must be supplied to the procedure so that the relief displacement at differing elevations can be accounted for. This process is known as **Ortho-rectification**. Used for mountainous terrains
- 7) **Level 3B** Mosaic Level 3A products to cover larger area

ERROR SOURCES



Error Sources are of TWO types

1) Internal Sources

- Introduced by the remote sensing system
- Generally systematic (predictable) and may be identified and then corrected based on prelaunch or in-flight calibration measurements

2) External Sources

- Introduced by phenomena that vary in nature through space and time
- Due to atmosphere, terrain elevation, slope, and aspect
- corrected by relating empirical ground observations

RADIOMETRIC PRE PROCESSING

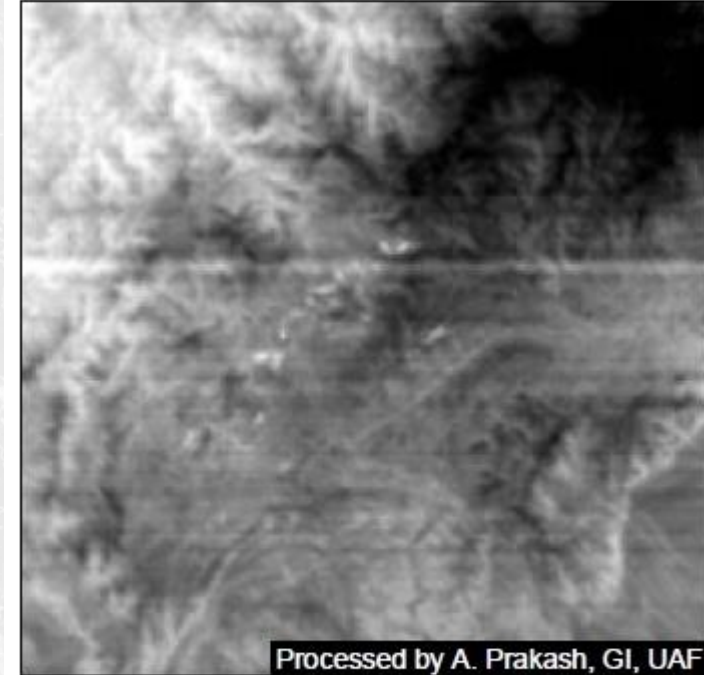
Types of Radiometric Correction

- Striping : (partially) missing lines
- Illumination and view angle effects
- Sensor calibration
- Terrain effects
- Atmospheric correction

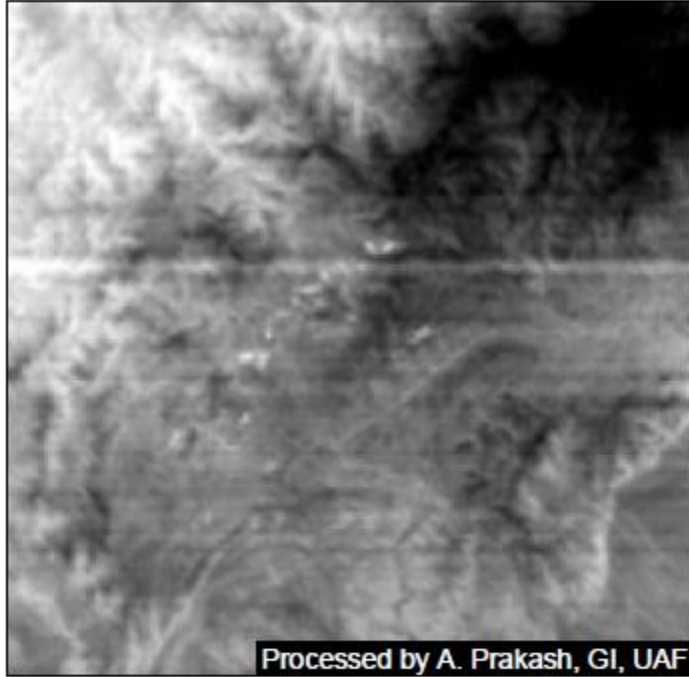
RADIOMETRIC PRE PROCESSING

Line or Column Striping

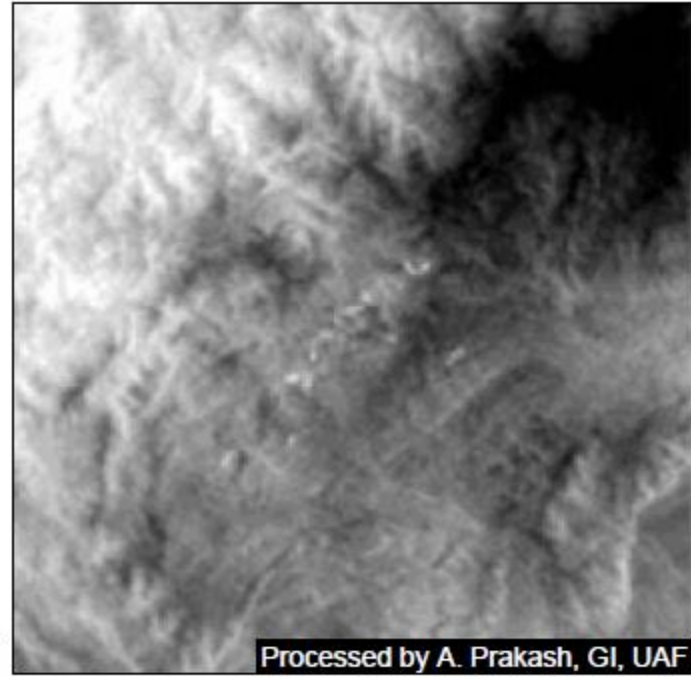
- Sometimes a detector does not fail completely, but simply goes out of radiometric adjustment
- The result would be an image with systematic, noticeable lines that are brighter than adjacent lines
- This is referred to as n-line striping
- To repair systematic n-line striping, it is first necessary to identify the miscalibrated scan lines in the scene.
- Every line and pixel in the scene recorded by the maladjusted detector may require a bias (additive or subtractive) correction or a more severe gain (multiplicative) correction



Striping



Striping



De-striped

RADIOMETRIC PRE PROCESSING

(Partially) missing lines

Due to Errors in

- ❑ sampling or scanning equipment
- ❑ transmission or recording of image data
- ❑ reproduction of the media containing the data

Methods of Correction

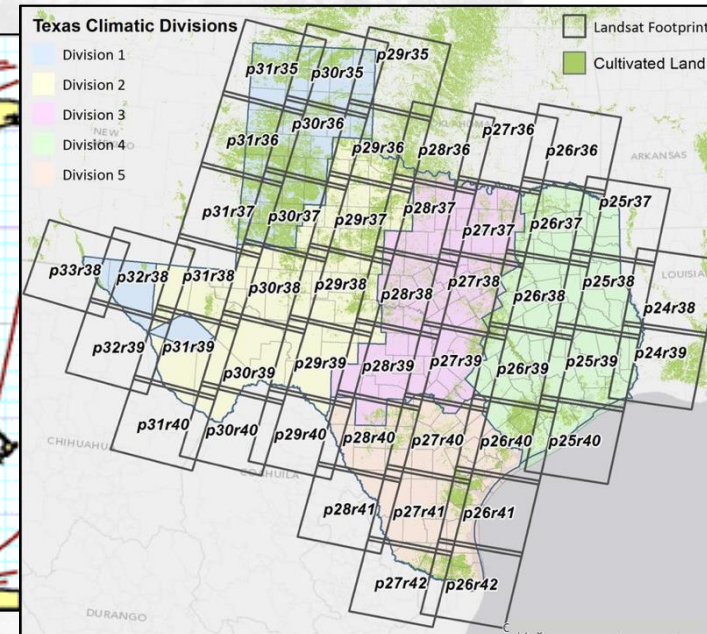
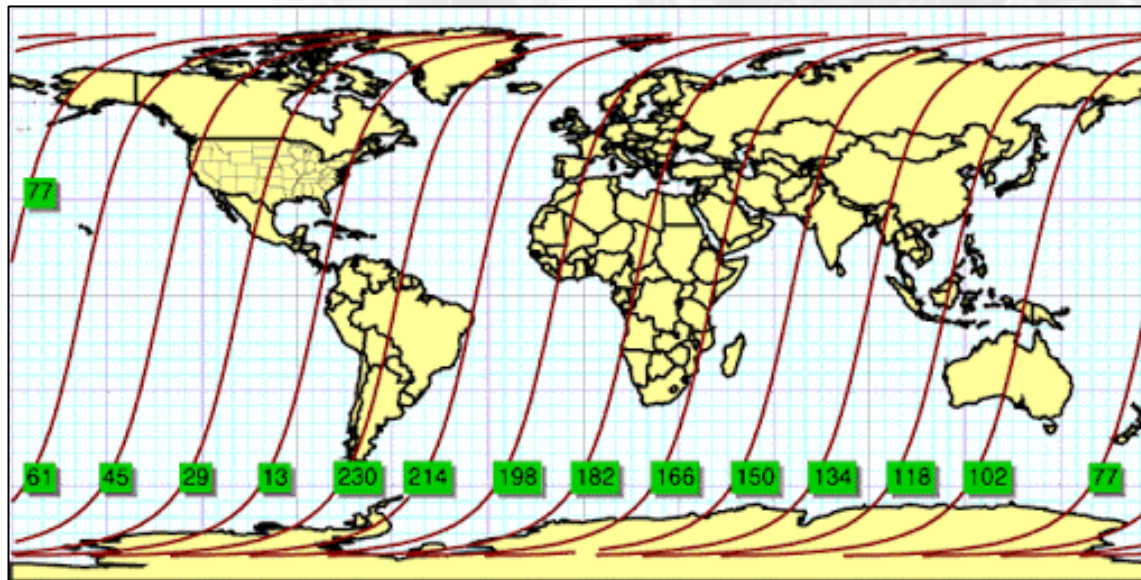
- ❑ interpolation using data from adjacent scan lines
- ❑ interpolation data at the same scan line from different spectral bands



IMAGE SCENE REPRESENTATION

LANDSAT

- Orbit paths are numbered westward
- Path 001 -- > Through eastern Greenland and south America
- Row refers to the latitudinal center line of a frame of imagery
- Rows are numbered southward beginning at 80 deg, N latitude
- Latitude with row 60 is closest to the equator
- Combination of path and row numbers identifies the scene center



GEOMETRIC CORRECTIONS

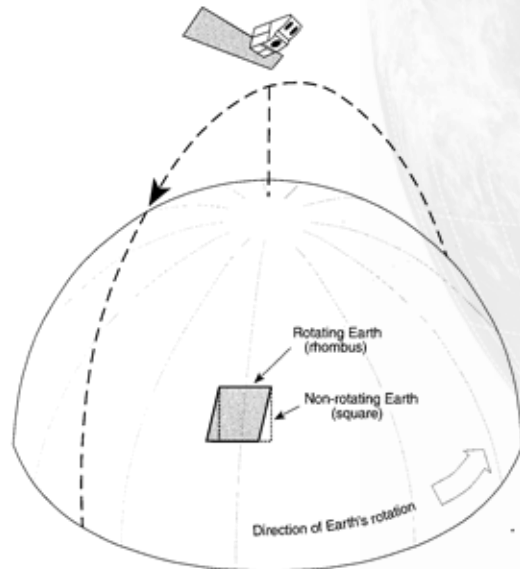
- Remotely sensed images are distorted by both the curvatures of the Earth and the sensor being used
- The process of shifting pixel locations to remove distortion is known as rectification or **georectification**
- The process of assigning geographic coordinates to an image is known as **georeferencing**
- Georeferencing is followed by **map projection** (by representing the surface of the sphere on a plane)
- Geometric correction is necessary to preprocess remotely sensed data and remove geometric distortion so that individual pixels are in their proper planimetric (x, y) map locations
- Geometrically corrected imagery can be used to extract accurate distance, polygon area, and direction information

GEOMETRIC ERRORS IN RS

- Introduced by the remote sensing system itself or in combination with earth rotation and curvature effects (internal errors) OR by random movements of space craft (external errors)
- **Internal errors** are systematic (predictable) and can be easily corrected, as below
 - 1) Skew caused by earth rotation effects
 - 2) Scanning system induced variation in ground resolution cell size
 - 3) Scanning system one-dimensional relief system
 - 4) Scanning system tangential scale distortion

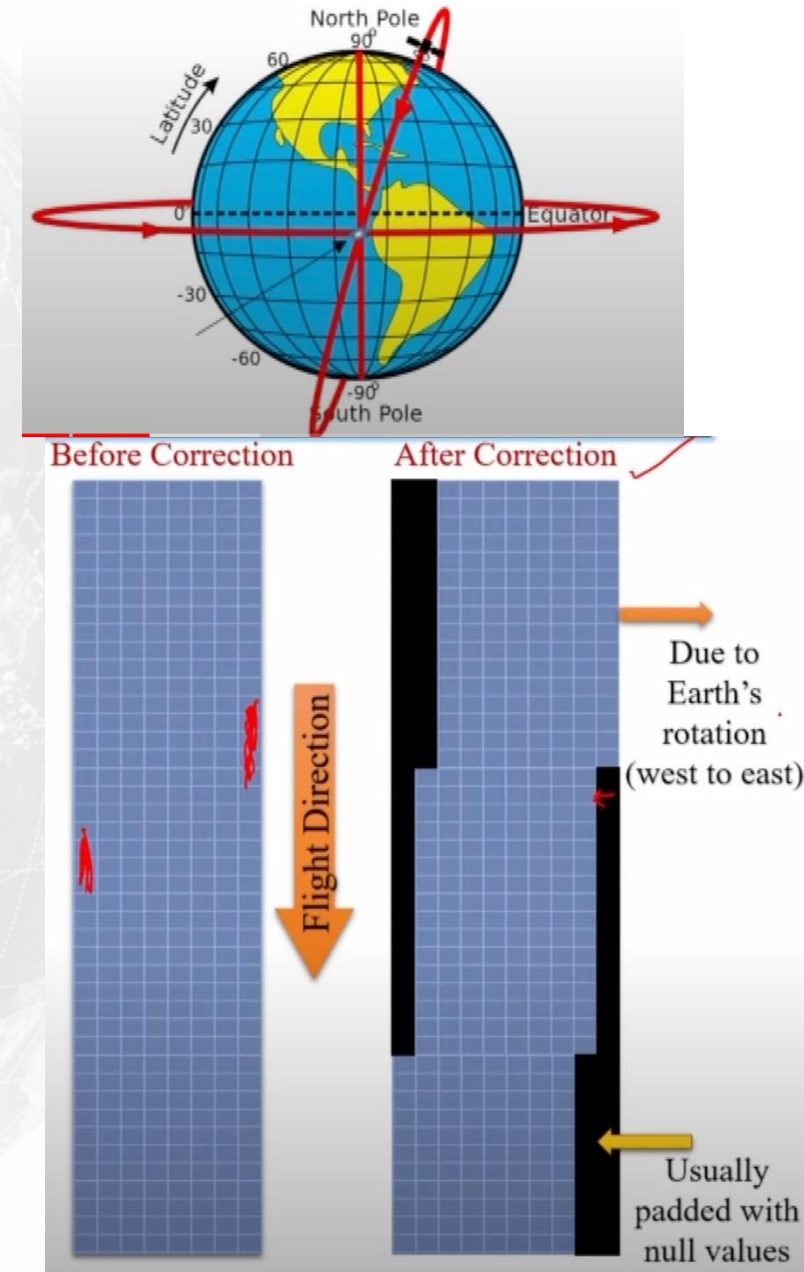
1. Earth Rotation Effects

- Land observation (sun synchronous) satellites have fixed orbits – North to south, where as earth rotation is west to east
- The interaction of these two skews the geometry of the image collected
- While the matrix (raster) may look correct, it actually contains systematic geometric distortion caused by the angular velocity of the satellite in its descending orbital path in conjunction with the surface velocity of the Earth as it rotates on its axis while collecting a frame of imagery
- Adjusting the image to the west to compensate earth rotation is called de-skewing



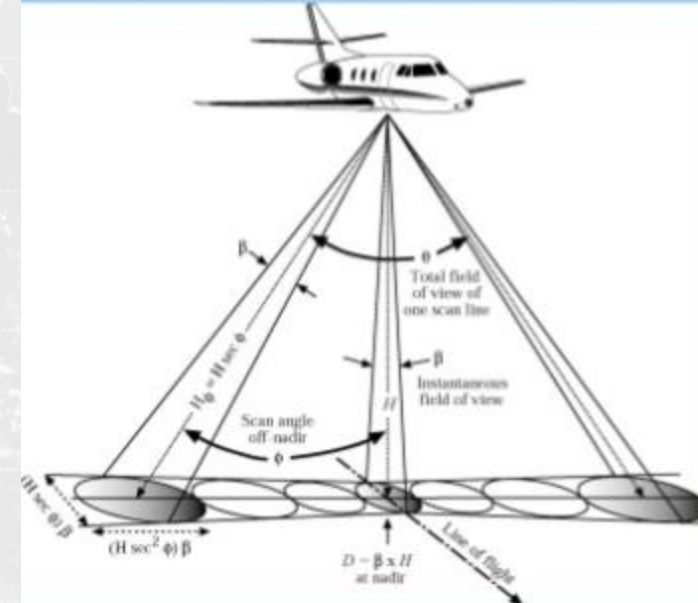
Scan Skew

- Satellite moving along its orbital track tries to scan the earth revolving with a surface velocity proportional to the latitude of the nadir.
- Results in shift in displacement of the last scan line in the image.



2. Variation in Ground cell resolution

- Happens along a single across-track scanning
- Instantaneous FoV define the image size
- Pixels off-nadir have semi major and semi minor axis that defines cell resolution

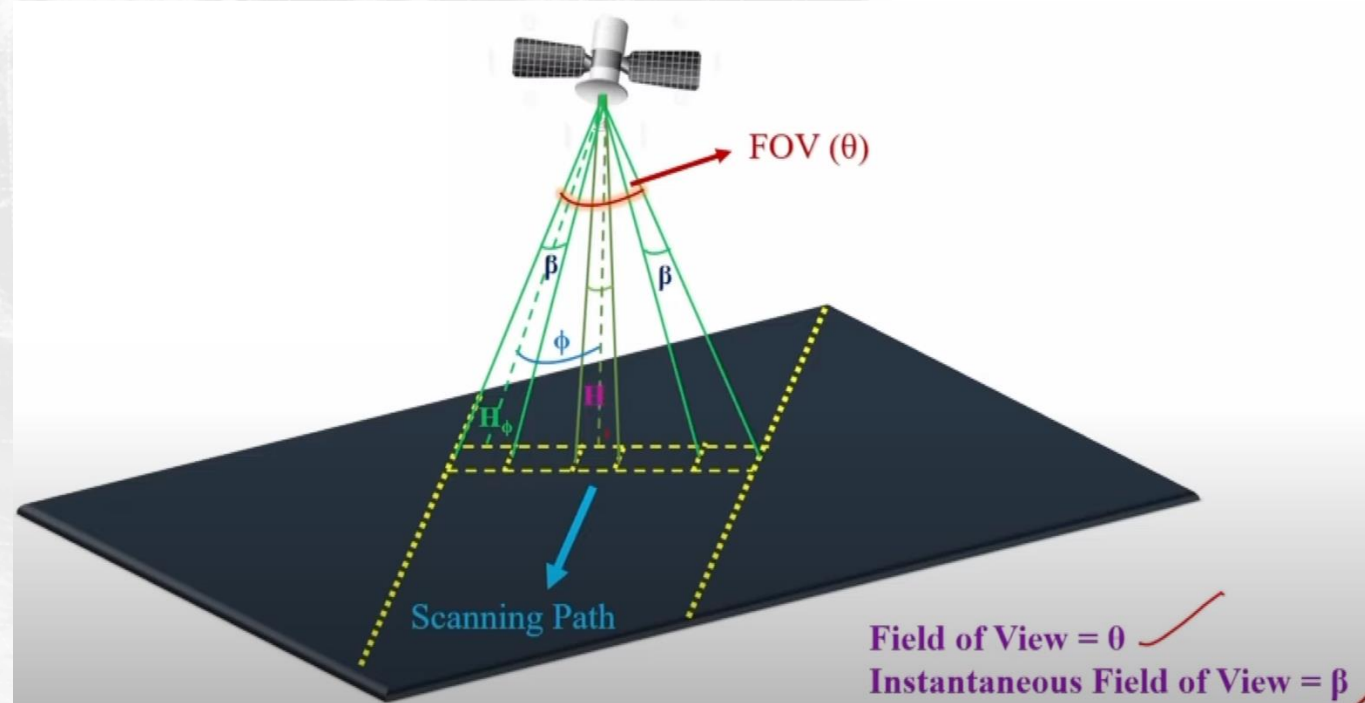


GROUND SWATH

- The width of the terrain strip remotely sensed by the system during one complete across-track sweep of the scanning mirror
- It is a function of angular field of view of the system (θ) and altitude of the sensor above ground level (H)
 - Ground swath width = $\tan(\theta/2) * H * 2$

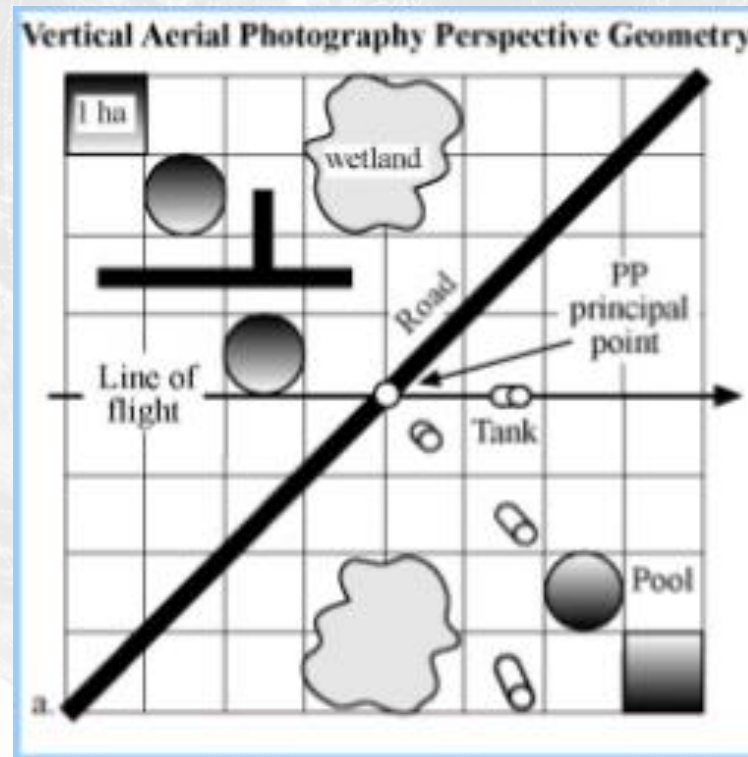
At Nadir: $D = \beta \times H$

Off-Nadir: $D = (H \sec \phi) \beta$



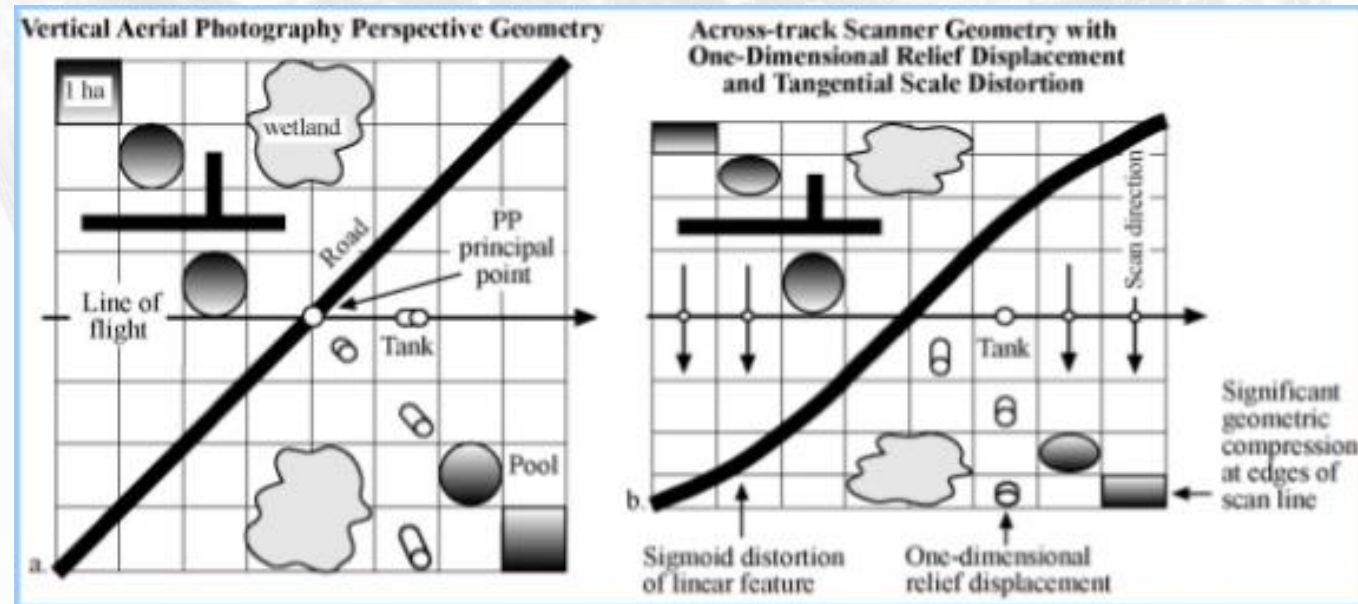
3. Variation in Relief displacement

- Relief displacement is the shift in an object's image position caused by its elevation above a particular datum
- Tangential scale distortion is the compression of image features at points away from the nadir
- Greater the height of object above local terrain, and greater is the distance from nadir (flight path), greater is relief displacement



4. Scale distortion

- Resulting from the mirror rotation at a constant rate
- Sensor scans shorter distance at nadir than at edge
- Objects near the nadir exhibit a proper shape and scale
- Greater the distance of ground-resolution cell from nadir, greater is the image scale compression – tangential scale distortion
- Linear features take S-shaped, sigmoid function

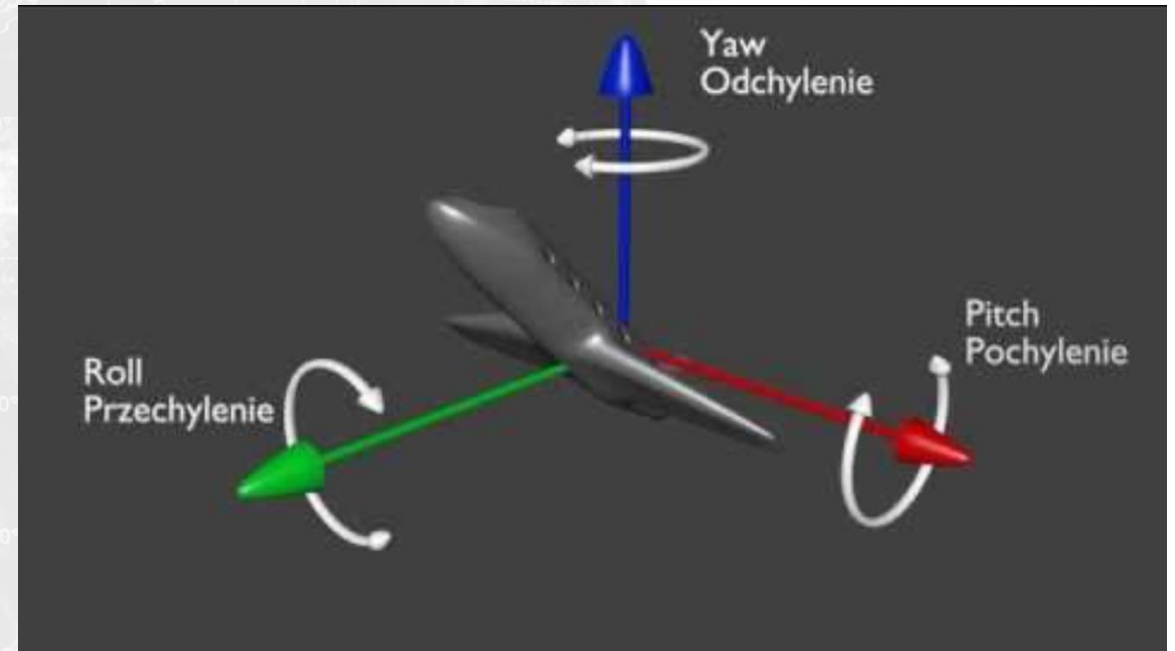
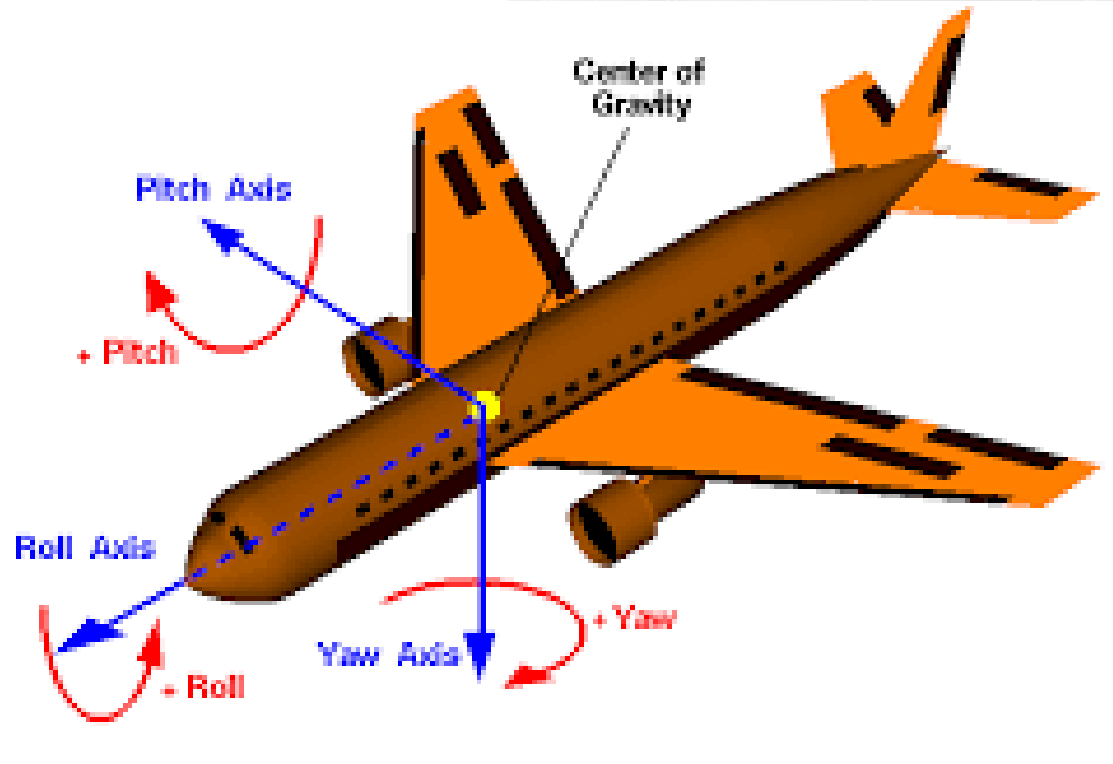


SATELLITE MOVEMENT AXIS

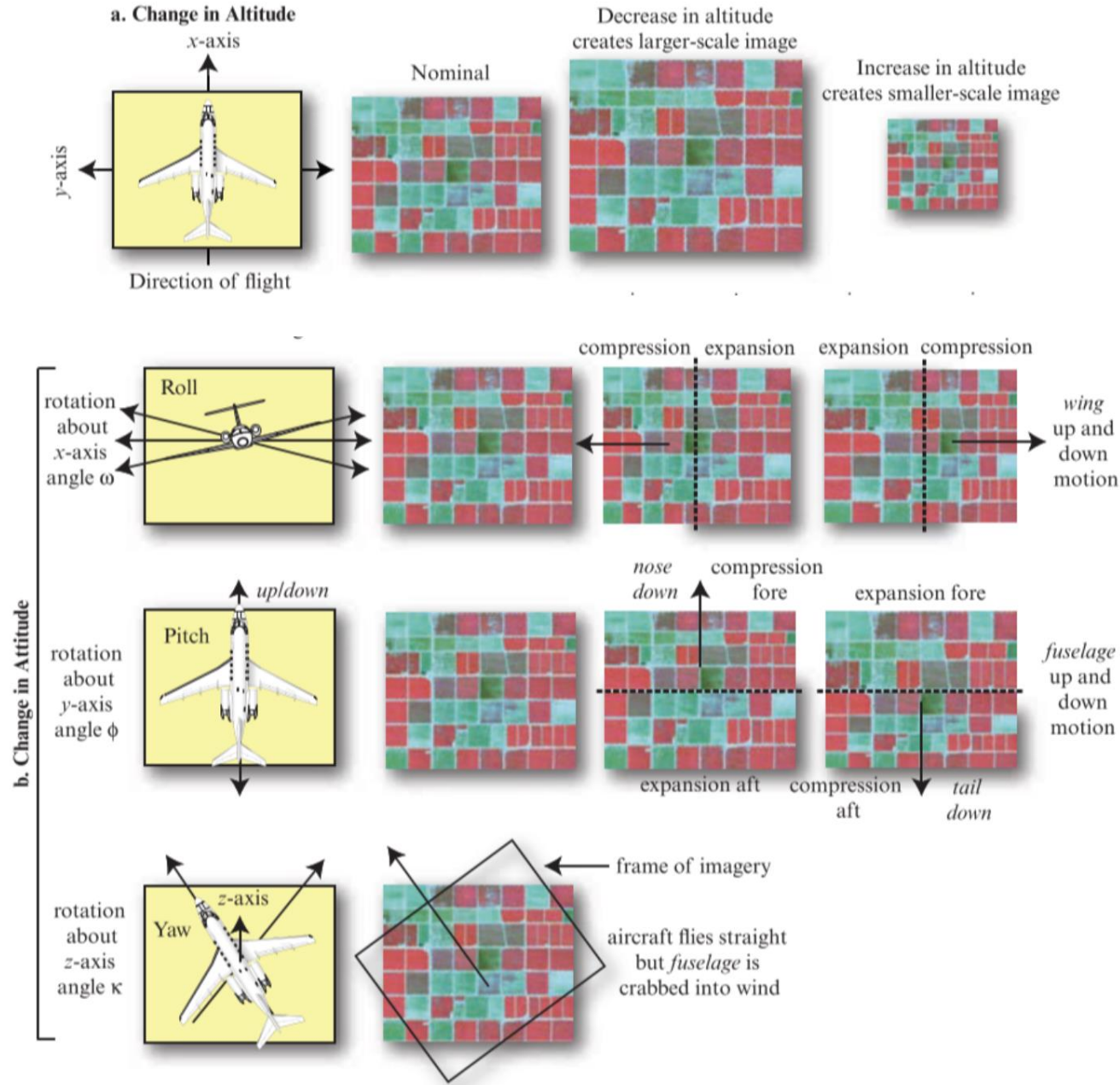
Rotation around the front-to-back axis is called **roll**

Rotation around the side-to-side axis is called **pitch**.

Rotation around the vertical axis is called **yaw**



EXTERNAL ERRORS



EXTERNAL ERRORS

- External geometric errors are usually introduced by phenomena that vary in nature through space and time
- Resulting from random movements by the aircraft (or spacecraft) at the exact time of data collection
 - *altitude changes, and/or*
 - *attitude changes (roll, pitch, and yaw)*
- Geometric distortions introduced by sensor system attitude (roll, pitch, and yaw) and/or altitude changes can be corrected using ground control points and appropriate mathematical models
- The image analyst must be able to obtain two distinct sets of coordinates associated with each GCP:
 - image coordinates specified in i rows and j columns, and
 - map coordinates (e.g., x, y measured in lat and long)

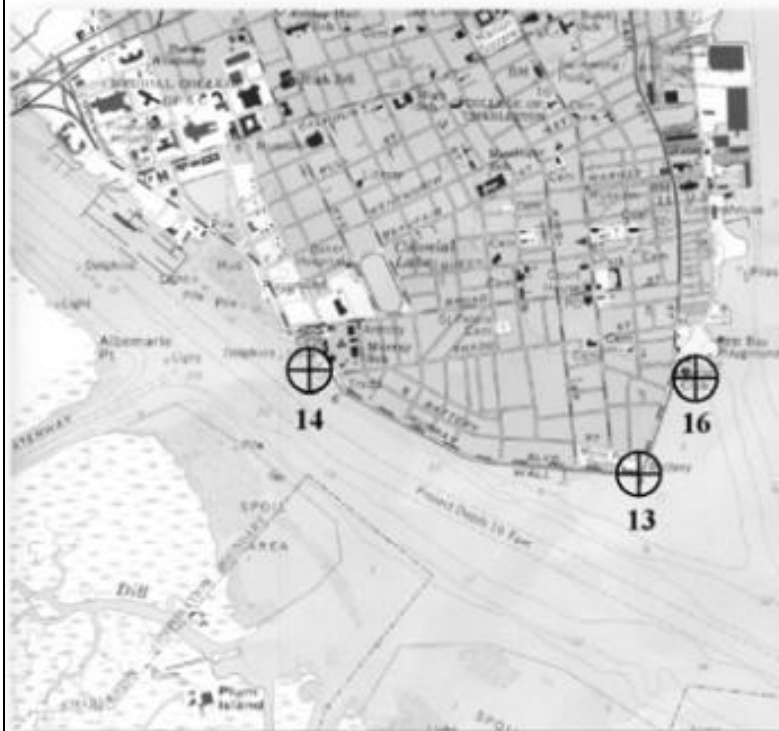
GROUND CONTROL POINTS

- Geometric distortions introduced by the sensor system can be corrected using ground control points (GCPs) and appropriate mathematical models
- The paired coordinates (i, j and x, y) from many GCPs (e.g., 20) can be modeled to derive geometric transformation coefficients.
- GCP → Location on the surface of earth that can be identified on imagery and located accurately on the map
- Geometrical rectification of an image ensure a standard datum and projection
- Can be done using:
 - Image to Map rectification
 - Image to Image registration

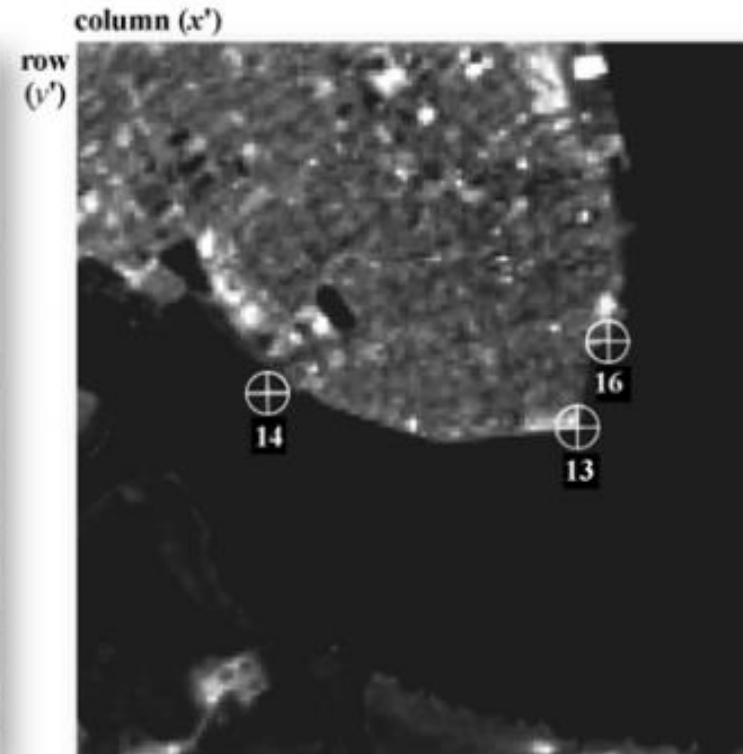
GROUND CONTROL POINTS

Image to Map Registration

Selecting Ground Control Points for Image-to-Map Rectification

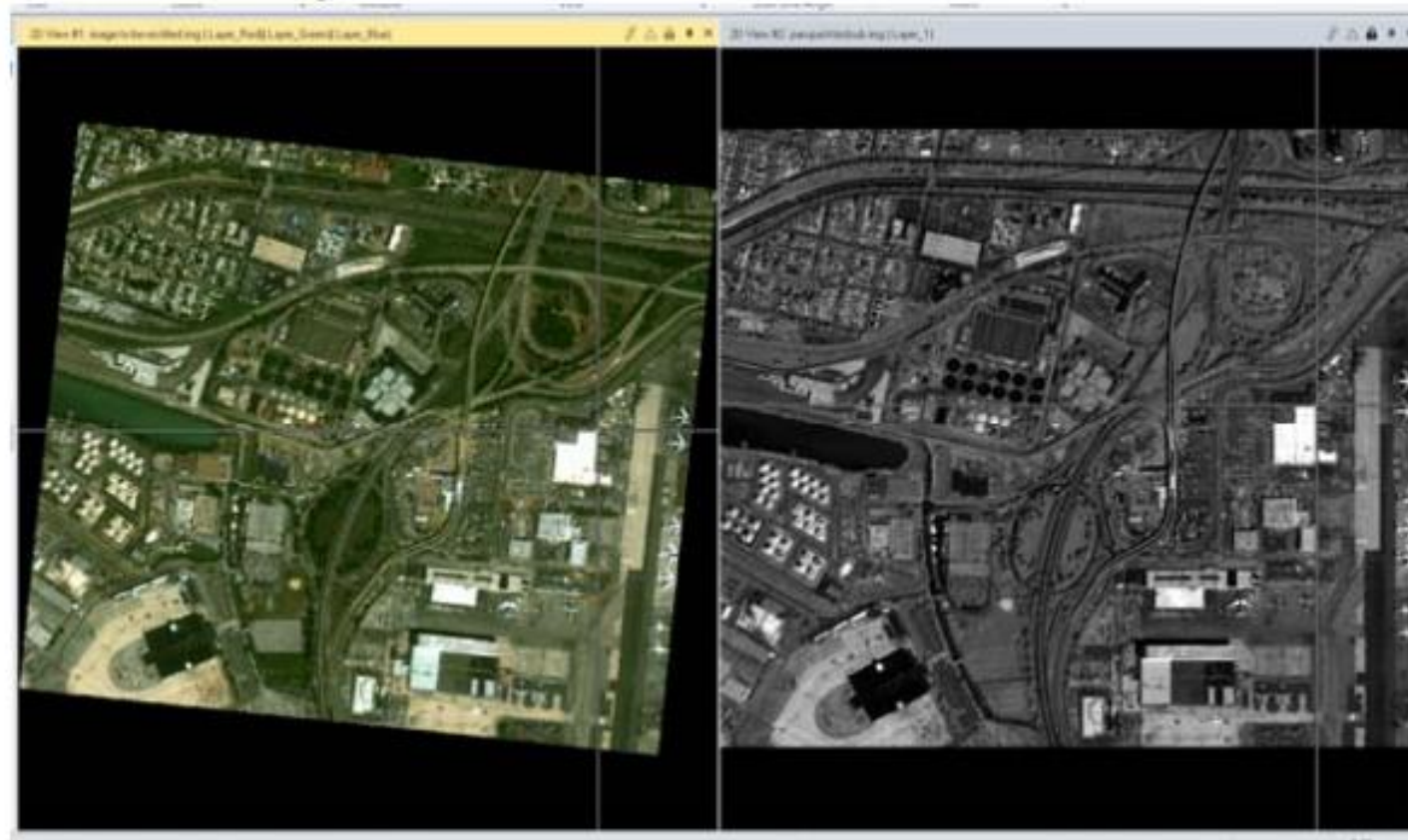


x
a. U. S. Geological Survey 7.5-minute 1:24,000-scale topographic map of Charleston, SC, with three ground control points identified.



b. Unrectified Landsat Thematic Mapper band 4 image obtained on November 9, 1982.

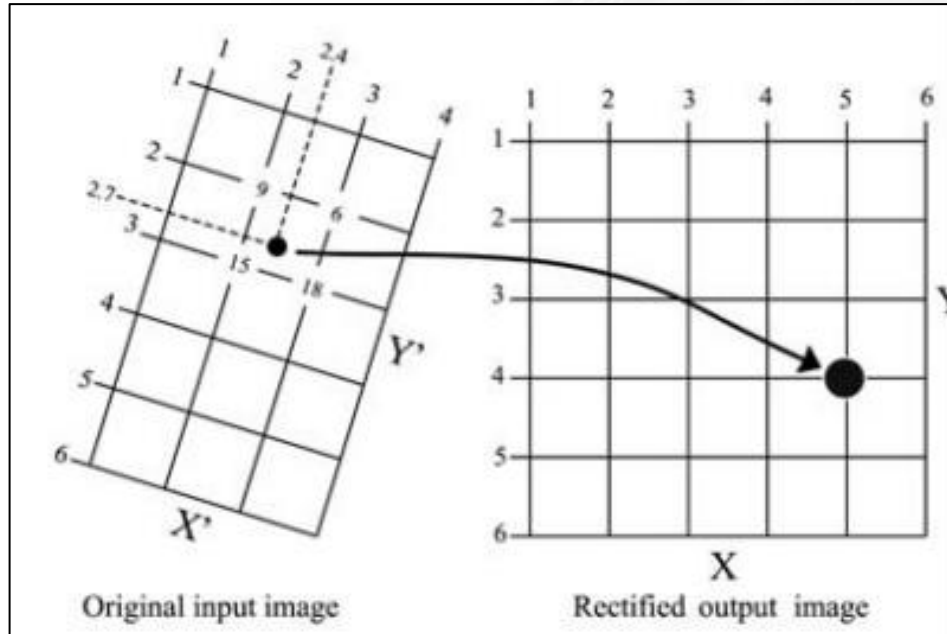
GROUND CONTROL POINTS



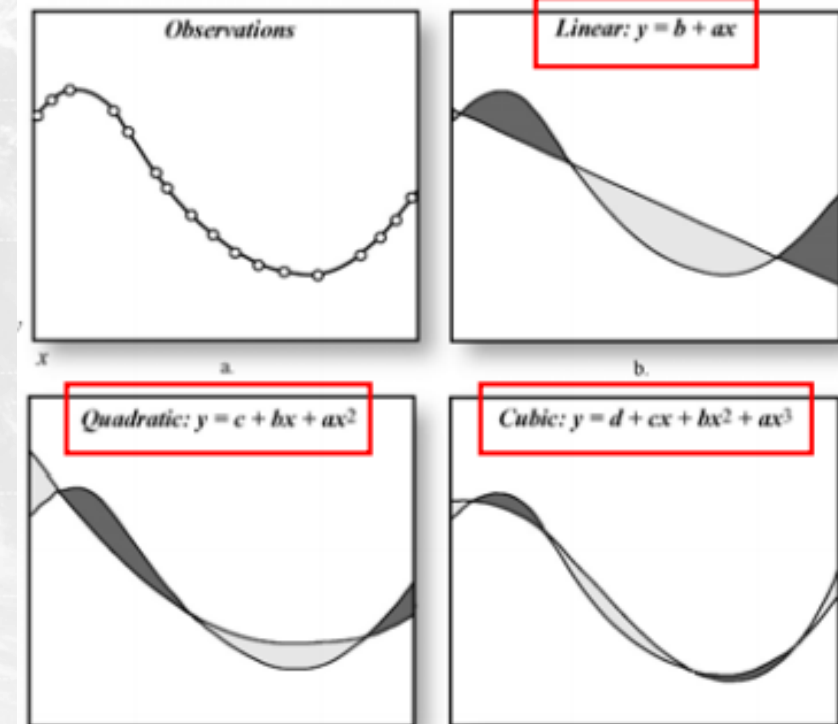
SPATIAL INTERPOLATION

- The geometric relationship between the input pixel coordinates (column and row; referred to as x' , y') and the associated map coordinates of this same point (X , Y) must be identified
- A number of GCP pairs are used to establish the nature of the geometric coordinate transformation that must be applied to rectify or fill every pixel in the output image (x , y) with a value from a pixel in the unrectified input image (x' , y')
- Polynomial equations are used to convert source file coordinates into the referencing map coordinates
- Depending upon the distortion in an image, the number of GCPs used, and their locations relative to one another varies

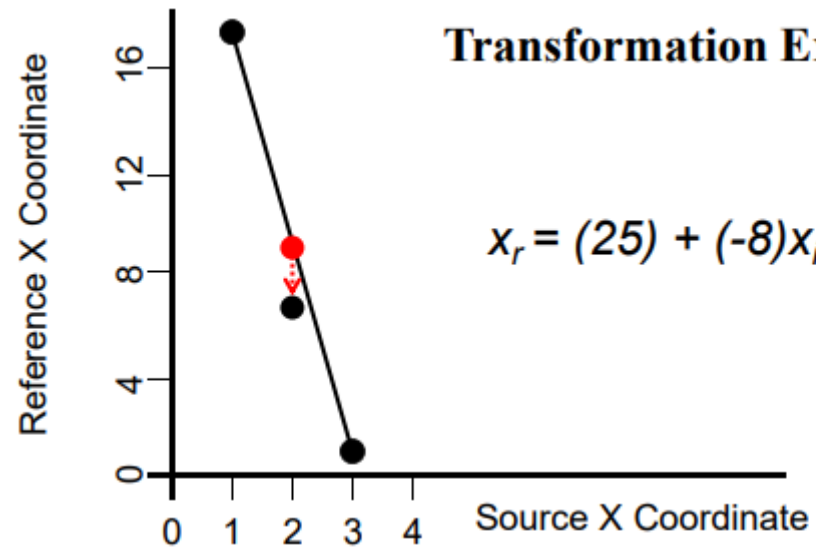
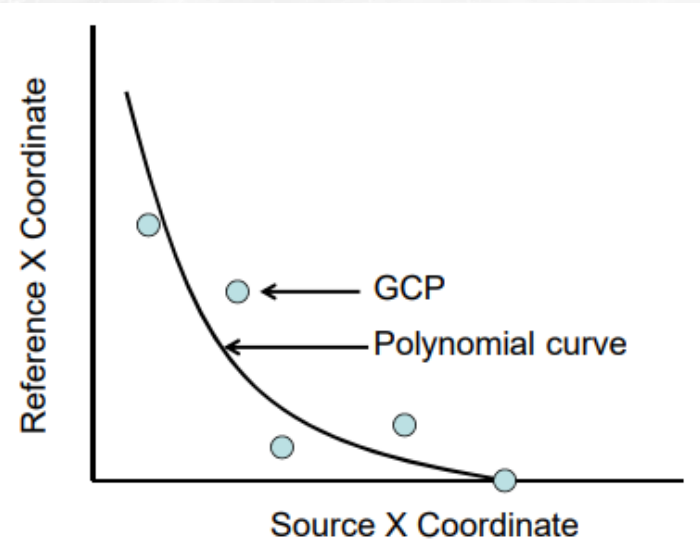
SPATIAL INTERPOLATION



- A transformation matrix is computed from the GCPs.
- The matrix consists of coefficients which are used in polynomial equations to convert the coordinates
- It is almost impossible to derive coefficients that produce no error.
- Every GCP influence the coefficients



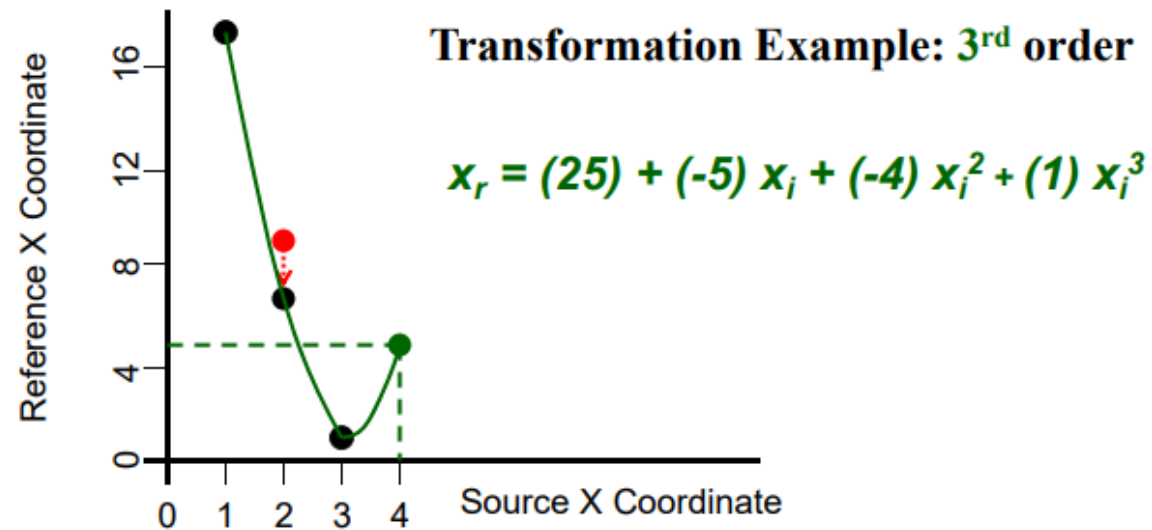
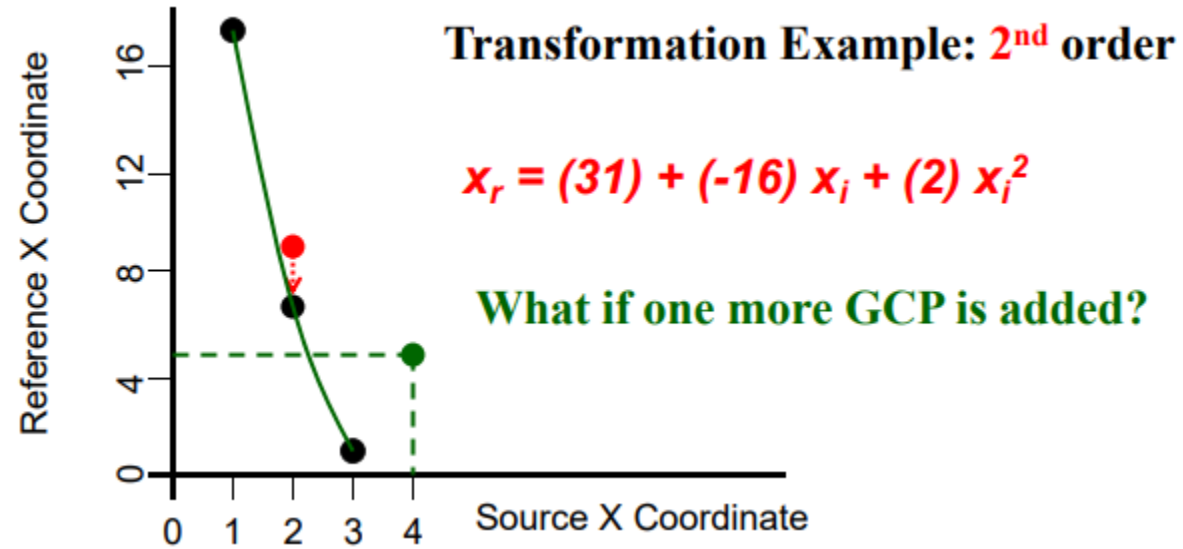
SPATIAL INTERPOLATION



Transformation Example: 1st order

$$x_r = (25) + (-8)x_i$$

SPATIAL INTERPOLATION



SPATIAL INTERPOLATION

- Higher orders of transformation can be used to correct more complicated types of distortion.
- However, to use a higher order of transformation, more GCP's are needed.
- The minimum number of points required to perform a transformation of order t equals

$$\frac{((t+1)(t+2))}{2}$$

Order of Transformation	Minimum GCPs Required
1	3
2	6
3	10
4	15
5	21
6	28
7	36

GEOMETRIC CORRECTION

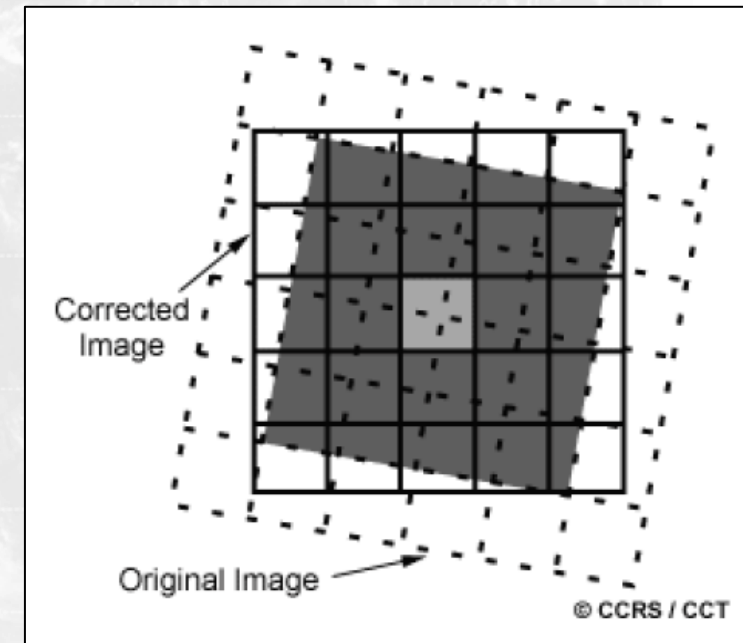
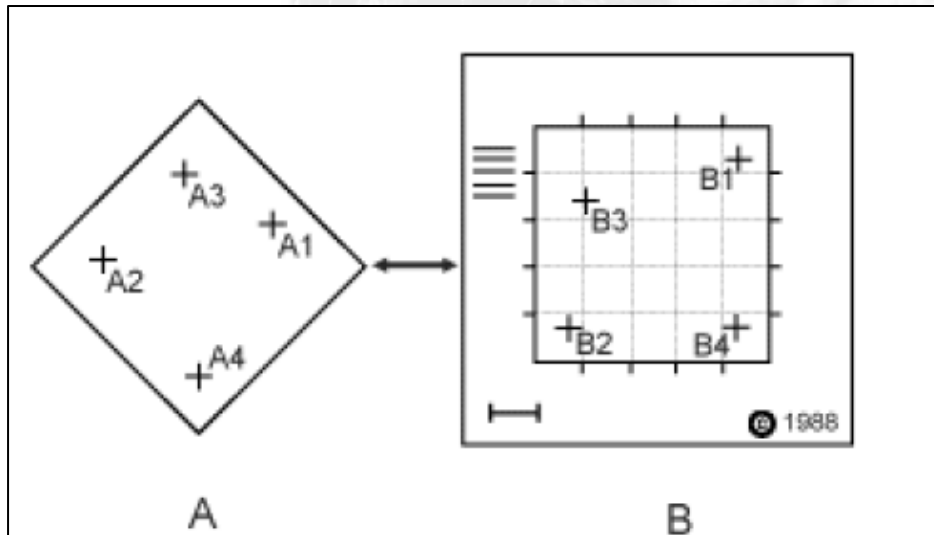
- Needed in preparation of planimetrically correct versions of aerial and satellite images
- Requires removal of positional effects of topographic relief to produce a positional accurate image
- Image registration is done using sampling techniques such as:
 - a) Nearest Neighborhood
 - b) Bilinear Interpolation
 - c) Cubic Convolution
- Resampling scales; rotates; translates; and performs related manipulations as necessary to bring the geometry of an image to match a particular reference image of desired properties
- Georeferencing is a resampling technique, where in matching is done not only to a reference image but also to reference points that corresponds to specific (known) locations on ground

GEOMETRIC CORRECTION

- Intensity interpolation involves the extraction of a brightness value from an x' , y' location in the original (distorted) input image and its relocation to the appropriate x , y coordinate location in the rectified output image
- There are several methods of brightness value (BV) intensity interpolation that can be applied
 1. nearest neighbor,
 2. bilinear interpolation, and
 3. cubic convolution.
- The practice is commonly referred to as resampling

GEOMETRIC CORRECTION

Geometric registration process involves identifying the image coordinates of several clearly discernible points, called ground control points (or GCPs), in the distorted image, and matching them to their true positions in ground coordinates (e.g. latitude, longitude)



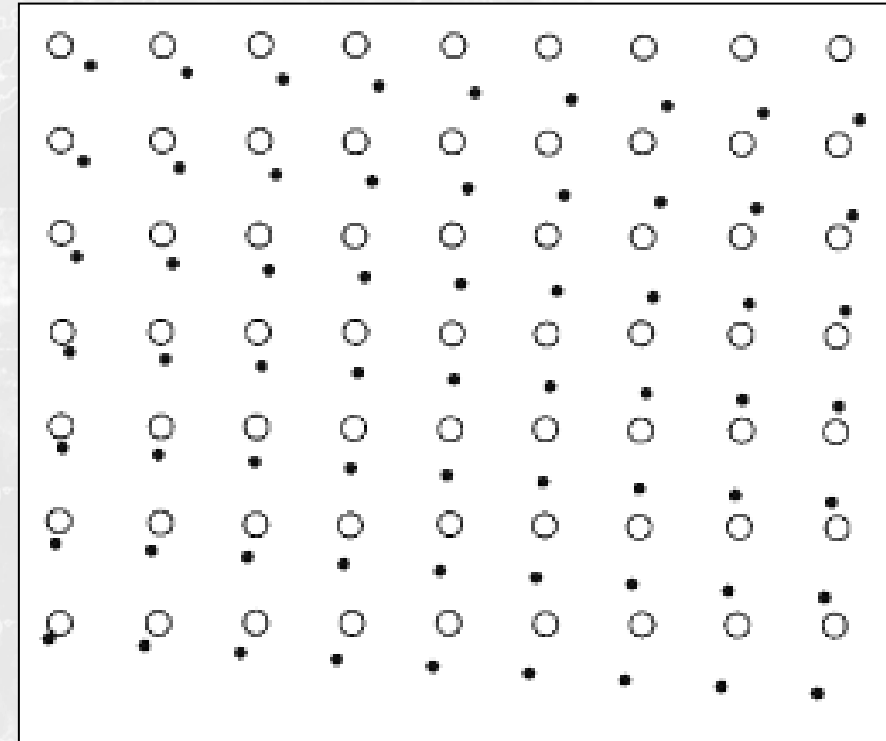
GEOMETRIC CORRECTION



GEOMETRIC CORRECTION

Resampling

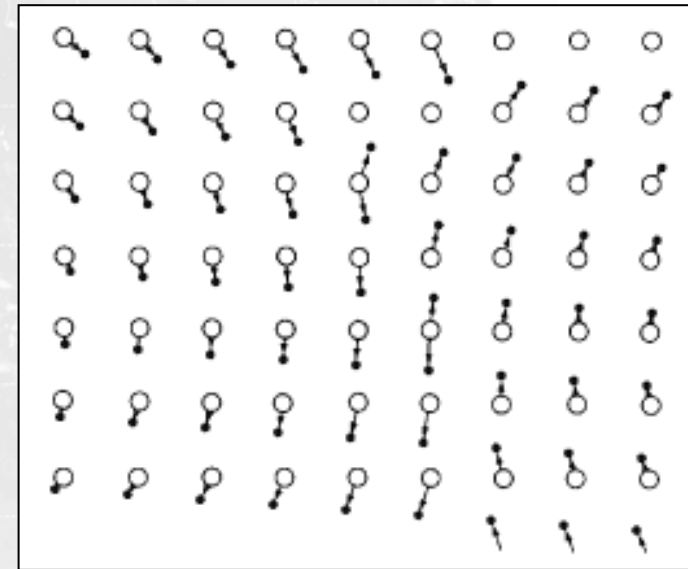
- Input Image → Array of open dots (Uncorrected Image)
- Output Image → Array of solid dots
- Locations of output pixels are derived from information provided by Ground Control Points (GCPs)
- Aim is to convert the input image for position and resample



GEOMETRIC CORRECTION

Nearest Neighbour

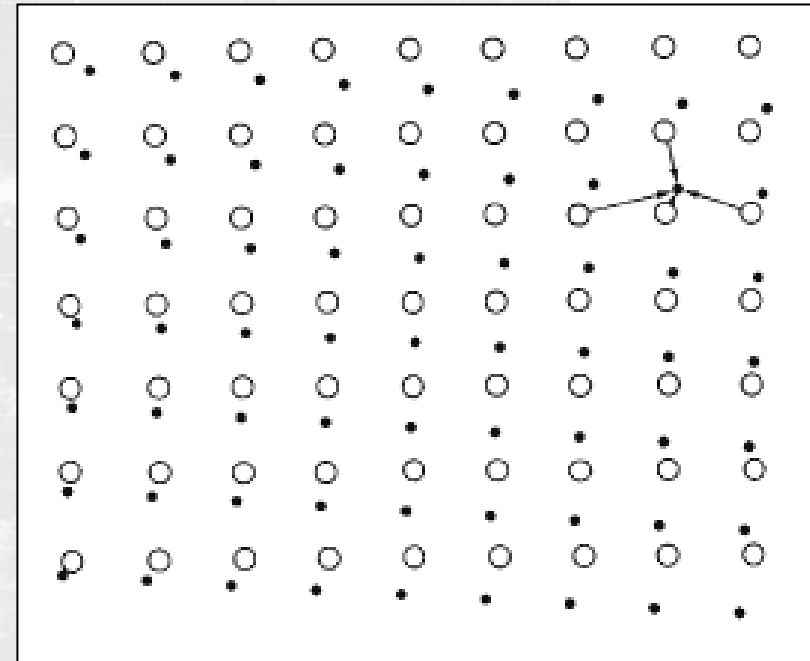
- Assign each “corrected” pixel the value from the nearest “uncorrected” pixel
- Simple to use
- Has the ability to preserve the original values of image
- May create noticeable positional errors
- The brightness value closest to the predicted x' , y' coordinate is assigned to the output x , y coordinate



GEOMETRIC CORRECTION

Bilinear Interpolation

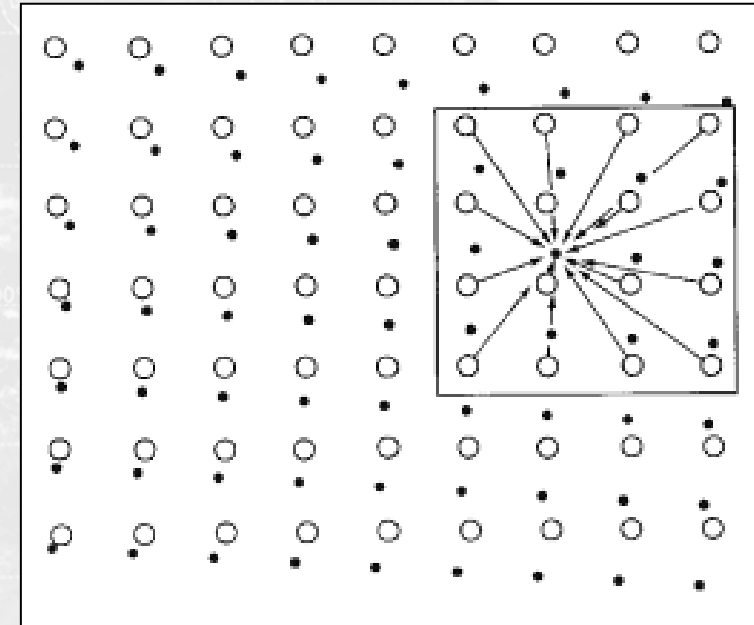
- Calculates a value for each output pixel based on a weighted average of the four nearest input pixels
- Nearer pixel values are given greater influence in calculating output pixel values
- Output image will not have the unnaturally blocky appearance of nearest neighbours
- Creates new pixel values. Brightness values in the input image are lost
- Since resampling is done on blocks of pixels, spatial resolution is lost

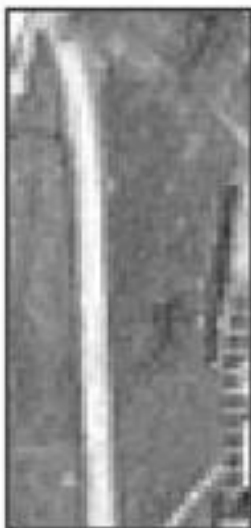
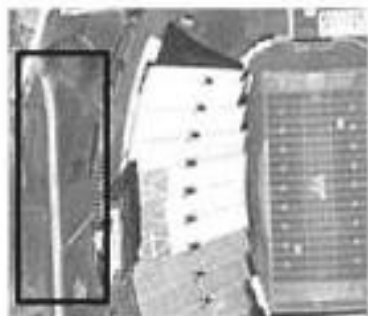


GEOMETRIC CORRECTION

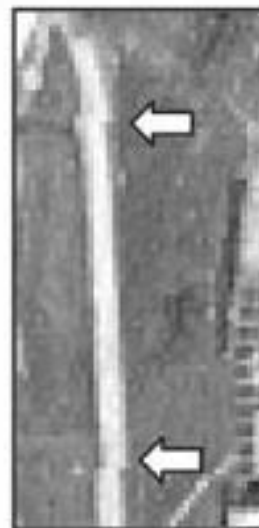
Cubic Convolution

- Most sophisticated, most complex, most widely used method
- Uses a weighted average of values within a neighbourhood that extends 2 pixels in each direction (a total of 16 pixels)
- Data is altered more than those of other methods
- Minimum number of GCPs needed is more
- Has natural appearance





**CUBIC
CONVOLUTION**



**NEAREST
NEIGHBOR**

